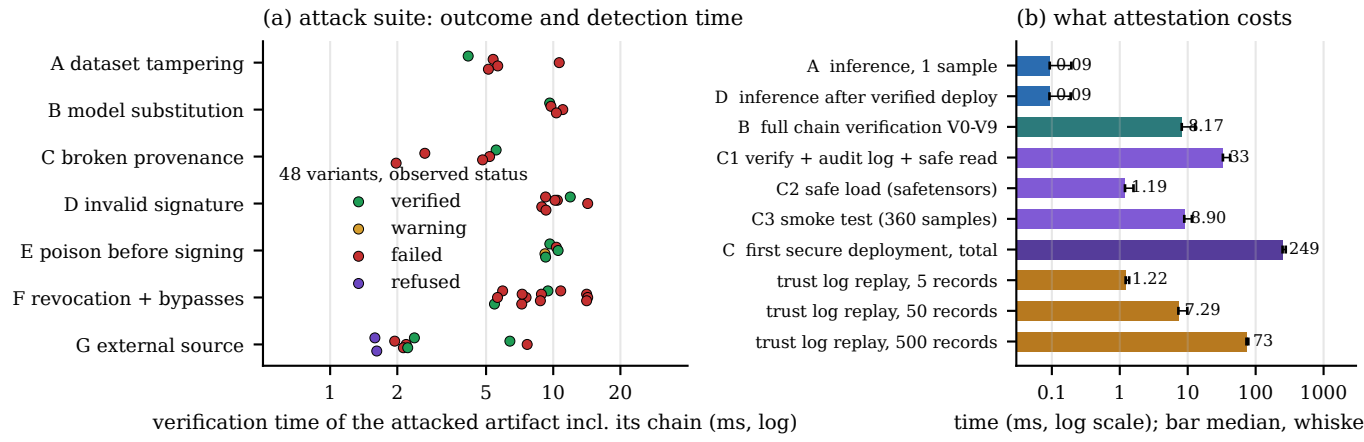
(a) attack suite: outcome and detection time
A dataset tampering
B model substitution
C broken provenance
D invalid signature
E poison before signing
F revocation + bypasses
G external source
48 variants, observed status
verified
warning
failed
refused
1
2
5
10
20
verification time of the attacked artifact incl. its chain (ms, log)
(b) what attestation costs
A inference, 1 sample
D inference after verified deploy
B full chain verification V0-V9
C1 verify + audit log + safe read
C2 safe load (safetensors)
C3 smoke test (360 samples)
C first secure deployment, total
trust log replay, 5 records
trust log replay, 50 records
trust log replay, 500 records
0.09
0.09
8.17
83
1.19
8.90
249
1.22
7.29
73
0.1
1
10
100
1000
time (ms, log scale); bar median, whiske